Alexandra Neagu

ngu.alexandra@gmail.com alexandra.neagu20@imperial.ac.uk +44 7443 976 277 (UK)
alexandra-neagu-743328116 neagualexa neagualexa.github.io

Hi, I am Alexandra — a second-year PhD researcher at Imperial College London studying how we can apply Large Language Models (LLMs) to support self-study in STEM, with publications at top AI+Education venues (ACM L@S 2026, EERN 2025). I worked for nine months at AWS Professional Services shipping serverless ML systems for internal and external customers.

Education

PhD in Applied AI in Education – Imperial College London *2024 – 2028*

- Home student, supervised by Dr Peter B. Johnson & Dr Rhodri Nelson.
- **Research question:** *"To what extent can existing GenAI tools provide effective dialogic feedback while aligning with educational values and ensuring domain-specific technical accuracy?"*
- Designing and evaluating LLM-powered dialogic feedback on Lambda Feedback, a self-study platform used across all faculties of Imperial; mixed-methods analysis of real-world student interactions at scale (LLM-as-judge, unsupervised clustering) validated against human-rated rubrics.
- Graduate Teaching Assistant: Self-Organised Multi-Agent Systems, ML for Data Science.

MEng Electronic and Information Engineering – Imperial College London *2020 – 2024*

- First Class Honours; Imperial College President's Undergraduate Scholarship for academic excellence.
- **Final Year Project:** *"Enhancing Puzzle-Solving with User-Tailored AI Assistance"* (supervised by Dr Nicole Salomons): controlled user study (n=25) measuring whether skill-tailored LLM hints improve novice learning on Nonogram puzzles. **Finding:** LLM assistance improved task efficiency but not learning gains.
- Relevant modules: Human-Centred Robotics, Self-Organised Multi-Agent Systems, Machine Learning, Privacy Engineering (Anonymisation, Differential Privacy), Computer Vision and Robotics (SLAM).

International Baccalaureate Diploma – Sevenoaks School, Kent *2018 – 2020*

- HL: Mathematics (7), Physics (7), Geography (6); SL: English Literature (5), German B (5), Mandarin B (6).

Publications

Neagu, A., Messer, M., Johnson, P.B. and Nelson, R., 2026. "'How Do I. . . ?': Procedural Questions Predominate Student-LLM Chatbot Conversations". *arXiv preprint*.

Sölch, M., **Neagu, A.**, Messer, M., Johnson, P., Kortemeyer, G., Ng, S.S.H., Lim, F.S. and Krusche, S., 2026. "μEd API: Towards a Shared API for Education Microservices". *ACM Conference on Learning@Scale 2026*.

Neagu, A., Johnson, P.B. and Nelson, R., 2025. "Chatbots for Dialogic Feedback during Self-Study: The Importance of Contextual Information". *EERN 2025 Symposium*.

Workshops & Tutorials

Build and Deploy Microservices for Automated Feedback — half-day tutorial, co-organiser with P.B. Johnson, M. Messer, F.S. Lim. *Festival of Learning June 2026, Seoul*.

PEAF 2026: Pedagogical Evaluation of Automated Feedback — full-day workshop, co-organiser with M. Messer, P.B. Johnson, C. Kandiko Howson, J. Savelka, S. Woodhead. *Festival of Learning June 2026, Seoul*.

Research Experience

SEFI Summer School on Engineering Education Research – Aalborg University *May 2026*

- Selected for a 1-week course hosted by the Aalborg UNESCO Centre for Problem-Based Learning in Engineering Science and Sustainability, Denmark, on connecting research with teaching and interdisciplinary collaboration.

LearnLab Summer School – Carnegie Mellon University *Aug 2025*

- Selected for a 1-week course on technology-enhanced learning and intelligent tutoring systems (ITS), Pittsburgh, USA.
- Trained a simulated student under the Knowledge-Learning-Instruction (KLI) framework to acquire politeness as knowledge components via interaction with an ITS tutor in a Chinese-learning task.

Industry Experience

Professional Services Industrial Placement – Amazon Web Services UK *Apr – Sept 2023*

- **Text-to-Speech pipeline proposal on AWS Polly (Emirates iXR use case):** designed and prototyped an end-to-end serverless TTS pipeline to generate audio for 3D avatars in employee training scenarios; packaged as a reusable Terraform deployment with a flexible UI, with resiliency, security, and high availability as core design constraints.
- **Global Monitoring & Alerting for the global EC2 fleet of AWS's virtual-desktop service**: built the back-end automation, QuickSight dashboards, and alarming system tracking metrics like active users, time-on-instance, CPU anomalies, idle instances — enabling the service team to right-size the fleet and flag unused instances or suspicious activity.
- Presented technical decisions in architecture reviews with senior consultants across EMEA and the US.

Professional Services Summer Intern – Amazon Web Services UK *Jul – Sept 2022*

- Migrated AWS's internal Excel-based Landing Zone Assessment onto the AWS Assessment Tool platform; shift allowed the offering from internal-only to customer- and partner-facing, enabling broader adoption as a standard practice.
- Cut consultant assessment time by **~20%** (~80h workflow) via automated report generation; scoped and requested platform features (e.g. weighted scoring) and implemented custom UI components under the new tool's constraints.

Skills & Certifications

Human-Centred ML & AI: Large Language Models, prompt engineering, LLM-as-judge & human evaluation, qualitative analysis at scale, user studies, PyTorch, agentic workflows.

Programming: Python, TypeScript, SQL, GraphQL, C#, Bash, Git, Unity.

Cloud & MLOps: AWS (Certified Machine Learning Speciality, Solutions Architect Associate, Cloud Practitioner), Google Cloud Platform, Docker, Pulumi, Terraform.

Languages: English (native), Romanian (native), German (B1-B2), Mandarin (HSK 4-5), French (A1-A2).

Selected Projects *(see more on my personal page)*

Avatour – VR-mediated Robot Telepresence **[University Project]** — *Jan – Mar 2024*

- **Outcome:** ran a controlled user study across 2D and VR interfaces on operating a Boston Dynamics SPOT robot for exploring restricted real-world locations; findings contradicted our hypothesis: prior user experience, not modality, dominated trust and immersion with the teleoperation interface.
- **Skills:** user study design, mixed-methods HCI evaluation, human-centered interface design.

CharIOT – Cleanroom Monitoring System **[University Project]** — *Jan – Mar 2023*

- **Outcome:** distributed IoT system enabling cleanroom researchers to detect environmental failures (temperature, pressure, humidity, particulates) in real time, with user-defined alarm thresholds synchronised across devices.
- **Skills:** distributed systems design, real-time sensor data pipelines, cloud-edge integration.

ZeroToOne – AR/VR Circuit Design Simulator **[Hackathon]** — *Feb 2022*

- **Outcome:** an AR/VR Android app delivering step-by-step tutor flow for remote hardware lab circuit assembly.
- **Skills:** drove idea-to-MVP process, prototyping under time & cost constraints, AR/VR interaction design for novice users.

External Courses

- Foundations of Engineering Education, University at Buffalo, NY (online, 12 weeks). — *2025*
- Nvidia: Accelerating CUDA C++ applications with multiple GPUs (2 days). — *2024*
- Computer Vision Course, Machine Learning University on AWS SageMaker (online, 3 days). — *2023*
- AWS Certified SysOps Administrator Associate, Udemy (online, 30 hrs). — *2023*

Volunteering & Hobbies

Vice President Industry, EESoc – Imperial College London — *2021 – 2023*

- Built partnerships between companies and the Electrical Engineering Society; led the organisation of one of Imperial's largest student-led Careers Fairs and Intro-to-Industry Events.

Hackathon Projects — *ongoing*

- Participated or volunteered at hackathons every year since 2021, building projects across various domains and technologies. (ICHack 2021-2026, AWS Interns Hack 2023, WSET Hack-her-thon 2021-2022).

Travel & Cross-Cultural Exchange — *ongoing*

- Travel across Europe, USA, China, Peru, India, Japan, Madagascar, Costa Rica, Tanzania, and Egypt — a core part of who I am: I love experiencing different cultures and the discrepancies in how people approach life.